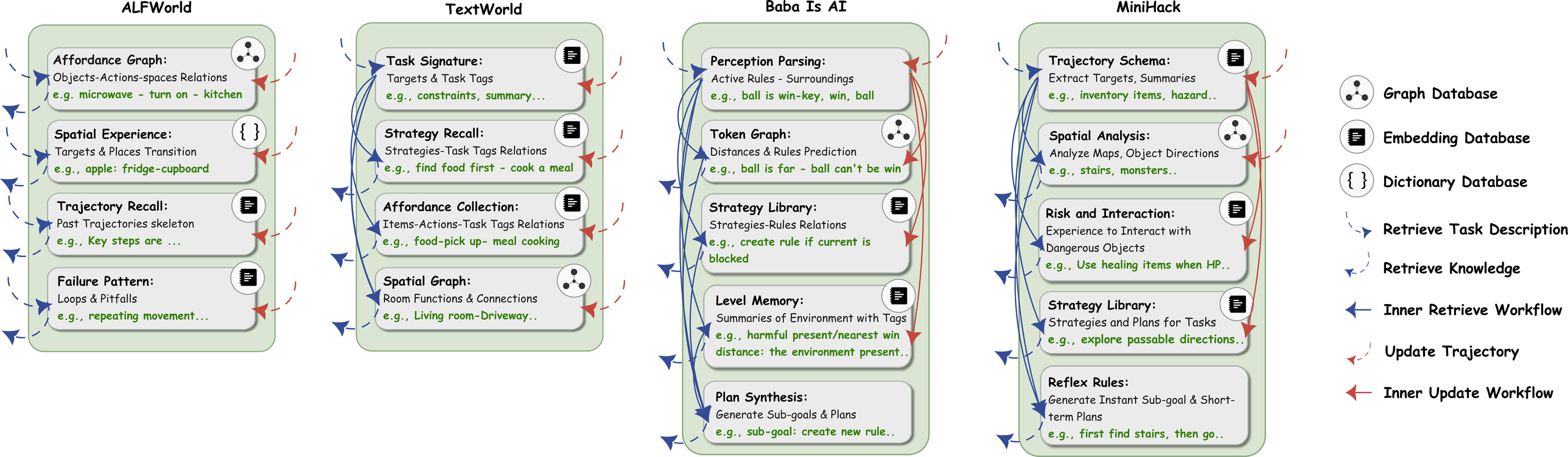
ALFWorld
Affordance Graph:
Objects-Actions-spaces Relations
e.g. microwave - turn on - kitchen
Spatial Experience:
Targets & Places Transition
e.g., apple: fridge-cupboard
Trajectory Recall:
Past Trajectories skeleton
e.g., Key steps are ...
Failure Pattern:
Loops & Pitfalls
e.g., repeating movement...
TextWorld
Task Signature:
Targets & Task Tags
e.g., constraints, summary...
Strategy Recall:
Strategies-Task Tags Relations
e.g., find food first - cook a meal
Affordance Collection:
Items-Actions-Task Tags Relations
e.g., food-pick up- meal cooking
Spatial Graph:
Room Functions & Connections
e.g., Living room-Driveway..
Baba Is AI
Perception Parsing:
Active Rules - Surroundings
e.g., ball is win-key, win, ball
Token Graph:
Distances & Rules Prediction
e.g., ball is far - ball can't be win
Strategy Library:
Strategies-Rules Relations
e.g., create rule if current is blocked
Level Memory:
Summaries of Environment with Tags
e.g., harmful present/nearest win distance: the environment present..
Plan Synthesis:
Generate Sub-goals & Plans
e.g., sub-goal: create new rule..
MiniHack
Trajectory Schema:
Extract Targets, Summaries
e.g., inventory items, hazard..
Spatial Analysis:
Analyze Maps, Object Directions
e.g., stairs, monsters..
Risk and Interaction:
Experience to Interact with Dangerous Objects
e.g., Use healing items when HP..
Strategy Library:
Strategies and Plans for Tasks
e.g., explore passable directions..
Reflex Rules:
Generate Instant Sub-goal & Short-term Plans
e.g., first find stairs, then go..
Graph Database
Embedding Database
Dictionary Database
Retrieve Task Description
Retrieve Knowledge
Inner Retrieve Workflow
Update Trajectory
Inner Update Workflow